

SALES FORECASTING & DEMAND INTELLIGENCE

Superstore Sales Performance Review • Fiscal Years 2015–2018

EXECUTIVE SUMMARY

Between 2015 and 2018, the business generated approximately **\$2.26 million** in total sales, led by the Technology category. Revenue follows a repeatable yearly rhythm: a quiet start each January and February, a steady middle stretch, and a reliable surge from September through December. We built and tested three forecasting models against real, already-known recent sales; the strongest performer landed within roughly **19%** of the actual figures. Technology stands out as the only category with clear projected growth, while Furniture, Office Supplies, and two of our four regions show early signs of softening worth a closer look. We also identified 15 weeks of unusual sales activity — mostly tied to holiday promotions or one-off large orders — and organized our 17 product lines into four demand groups, each calling for its own stocking approach.

KEY FINDINGS

- ◆ **Technology leads the business** at \$827,456 in sales (36.6% of the total), ahead of Furniture (\$728,659) and Office Supplies (\$705,422).
- ◆ **The East region is our steadiest performer**, growing 18.1% year over year on average with very little swing — the most dependable region for planning.
- ◆ **Shipping is fast and consistent everywhere**, averaging under 4 days company-wide with almost no variation by region (3.9–4.1 days).
- ◆ **September, November, and December** are consistently our three strongest months, every single year — a reliable pattern, not a coincidence.
- ◆ **Sales are stable enough to forecast with confidence**, and this recurring yearly cycle was built directly into our forecasting model.

3-MONTH SALES FORECAST

We tested three forecasting tools by asking each to predict three months of sales we already knew the answer to. Our machine-learning model (XGBoost) came closest overall, missing by about 19% on average. Our statistical model (SARIMA) came a close second and, unlike the others, also provides a likely range rather than a single number — useful for understanding how much uncertainty is involved. Based on that model, monthly sales should fall within the ranges below roughly 19 times out of 20:

Month	Most Likely Sales	Likely Range (95% confidence)
Month 1	~\$54,800	\$30,000 – \$79,500
Month 2	~\$88,400	\$63,700 – \$113,200
Month 3	~\$92,600	\$67,800 – \$117,300

These ranges are intentionally wide — an honest reflection of how much sales can swing month to month. The safest approach is to plan around the midpoint while keeping both ends in view.

NOTABLE SALES ANOMALIES

Two independent detection methods scanned four years of weekly sales for unusual activity. Only one week was flagged by both at once — our highest-confidence finding. The three most notable weeks:

Week	Sales	Likely Cause
22 Mar 2015	\$37,704	A one-off large order or regional promotion. Flagged by both checks — our most confident finding.
18 Nov 2018	\$30,572	Year-end holiday / Black Friday promotional period.
2 Dec 2018	\$35,999	Continued year-end holiday buying surge.

PRODUCT DEMAND GROUPS & STOCKING STRATEGY

Demand Group	Products	Recommended Approach
High Volume, Growing	Accessories, Binders, Chairs, Phones, Storage, Tables, Copiers	Keep strong safety stock with automatic reordering — these drive most of our revenue. Review monthly, since demand here swings the most. Copiers (479% growth) deserves individual attention.
Low Volume, Growing	Appliances, Art, Bookcases, Envelopes, Fasteners, Furnishings, Labels, Paper, Supplies	Low priority for capital today, but scale ordering up gradually as growth continues. Watch for any of these breaking into the high-volume group.
High Volume, Declining	Machines	Wind down large standing stock and shift to smaller, more frequent orders to avoid unsold inventory as demand falls.

BUSINESS RECOMMENDATIONS

- ◆ **Prioritize Technology** in next quarter's budget — it already drives over a third of all revenue and is the only category with clear projected near-term growth (+7.3%).
- ◆ **Build a formal pre-holiday stocking plan** for September through December — reliably our strongest months every year, and home to several of our largest sales spikes.
- ◆ **Tighten stock discipline on Machines** and keep a close watch on Furniture and Office Supplies — Machines is high-volume but declining and ties up capital at markdown risk, while the other two show early signs of near-term softening.

A note on limitations: this model learns from four years of past orders and assumes future demand will keep following the same seasonal and growth patterns. It cannot anticipate sudden shocks — a new competitor, a supply disruption, or an economic downturn — and is less accurate for smaller product lines with limited history. Treat these forecasts as a planning input to be reviewed monthly, not as a fully automatic ordering system.